CS246: Database Management Systems Lab

Week # 03 (1 Questions, 100 Marks)
Lab session: AL1
Held on: 20-Jan-2025 (Mon)
Lab Timings: 14:00 to 17:00 Hours Pages: 2
Submission time: 16:45 Hrs, 20-Jan-2025
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Question 1: (100 points)

Query implementation using C language. Given an origin and destination, you should give the list of flights routes (1) with one hop (2) with two hops. Use of any other programming language is not allowed and will lead to awarding 0 marks.

Task 01 (2 marks) Declare a structure `time` containing two elements namely

1. hrs `int`
2. mins `int`

Task 02 (5 marks) Declare a structure `record` containing five elements

1. Origin `char(3)`
2. Destination `char(3)`
3. Flight number `char(6)`
4. Departure time `struct time`
5. Arrival time `struct time`

Task 03 (3 marks) Read input file `week02-database.txt` of the following format into the `record` structure.

```
BOM HYD  6E5056  21:30 22:55
BOM GAU  6E228   08:40 11:30
...
```

Task 04 (80 marks) Read the origin and destination airports and compute and list

1. (30 marks) all two hop flight routes.
2. (20 marks) all two hop flights where time difference between arrival of incoming flight and departure of outgoing flight should be with in one hour or more
3. (30 marks) all three hop flight routes

Task 05 (10 marks) Output at the end of Task 04 should be written in a new file with the each line format identical to the input file format

Two Hop Example Given the origin=GAU and destination is MAA then two hop flights example is (without the constraint incoming flight time and departure time of out going flight):

```
6E5203 | GAU  | DEL
6E2006 | DEL  | MAA

6E5203 | GAU  | DEL
6E2217 | DEL  | MAA
...
```

Three Hop Example Given the origin=GAU and destination is MAA then three hop flights example is (without the constraint incoming flight time and departure time of out going flight):

```
6E5056 | GAU | HYD
6E7890 | HYD | BLR
6E4567 | BLR | MAA
```